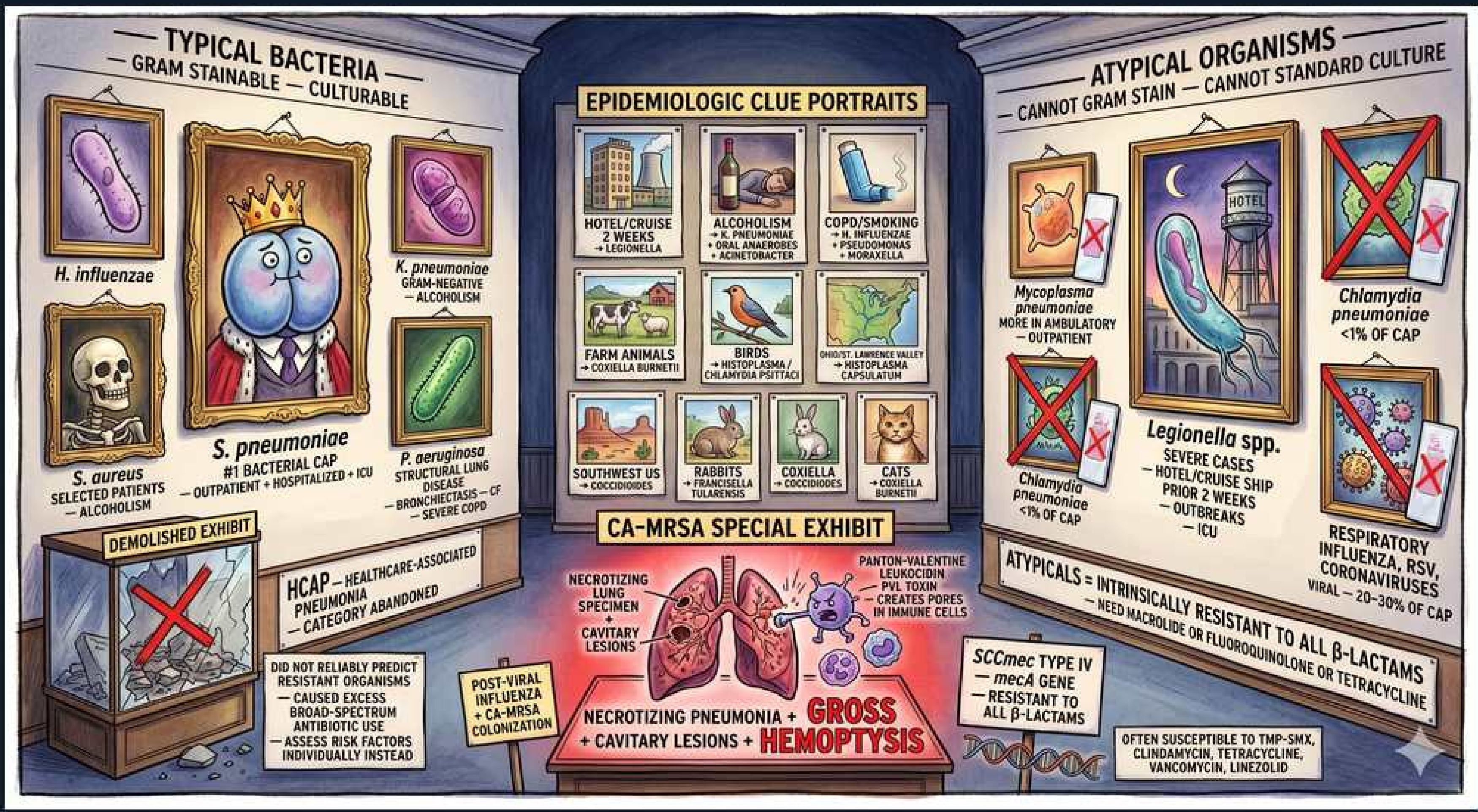
# Pneumonia Pathogens: Typical, Atypical & CA-MRSA



## 1. S. pneumoniae (King)

**WHAT YOU SEE**

Crowned king in framed portrait, gram-positive panel

**WHAT IT ENCODES**

#1 cause of CAP overall and the most common bacterial pneumonia. Lancet-shaped gram-positive diplococcus, alpha-hemolytic, optochin-sensitive, bile-soluble. Classic rusty-colored sputum and lobar consolidation. Treat with beta-lactam; pneumococcal vaccines (PCV/PPSV23) prevent invasive disease.

**BOARD TRAP**

Don't pick Klebsiella for the typical lobar CAP patient — Klebsiella gives currant-jelly sputum in alcoholics/aspiration, not rusty sputum.

## 2. H. influenzae

**WHAT YOU SEE**

Small rod portrait, top-left typical panel

**WHAT IT ENCODES**

Gram-negative coccobacillus; classic CAP in COPD/smokers and the elderly. Requires factors V and X on chocolate agar. Nontypeable strains cause most adult disease post-Hib vaccine. Covered by ceftriaxone or amoxicillin-clavulanate.

**BOARD TRAP**

H. flu is a COPD-exacerbation organism — don't reflexively assign every COPD pneumonia to Pseudomonas, which needs structural lung disease/bronchiectasis risk.

## 3. K. pneumoniae

**WHAT YOU SEE**

Capsule rod, alcoholism tag

**WHAT IT ENCODES**

Gram-negative rod causing aspiration pneumonia in alcoholics and diabetics. Currant-jelly sputum, thick mucoid capsule, upper-lobe cavitation, and bulging-fissure sign on CXR. Increasingly ESBL/carbapenem-resistant.

**BOARD TRAP**

Currant-jelly sputum = Klebsiella, NOT rusty sputum (S. pneumoniae) — the color/word is the discriminator.

## 4. S. aureus (skull)

**WHAT YOU SEE**

Skull, demolished exhibit, selected-patients tag

**WHAT IT ENCODES**

Causes post-influenza necrotizing pneumonia and pneumonia in IVDU. Gram-positive cocci in clusters, catalase and coagulase positive. Suspect after a flu prodrome with rapid cavitation and effusion.

**BOARD TRAP**

Necrotizing post-viral pneumonia points to S. aureus, not S. pneumoniae — the influenza prodrome is the clue.

## 5. P. aeruginosa

**WHAT YOU SEE**

Structural-lung-disease tag, bronchiectasis

**WHAT IT ENCODES**

Gram-negative rod in patients with structural lung disease (bronchiectasis, CF), neutropenia, or recent hospitalization. Oxidase-positive, grape-like odor, blue-green pigment. Needs antipseudomonal coverage (piperacillin-tazobactam, cefepime, ciprofloxacin).

**BOARD TRAP**

Reserve Pseudomonas coverage for structural lung disease/recent antibiotics — adding it to routine CAP is a common over-treatment trap.

## 6. Mycoplasma pneumoniae

**WHAT YOU SEE**

Crossed-out gram-stain frame, atypical panel

**WHAT IT ENCODES**

No cell wall so it cannot be gram-stained and is intrinsically resistant to all beta-lactams. Walking pneumonia in young adults, bullous myringitis, cold-agglutinin hemolysis. Treat with a macrolide or doxycycline.

**BOARD TRAP**

Atypicals are resistant to ALL beta-lactams (no cell wall) — picking ceftriaxone monotherapy for atypical CAP is wrong; you need a macrolide/doxy/fluoroquinolone.

## 7. Chlamydia pneumoniae

**WHAT YOU SEE**

Crossed gram-stain frame, <1% CAP tag

**WHAT IT ENCODES**

Obligate intracellular atypical causing mild CAP, pharyngitis-to-pneumonia, and hoarseness. Cannot gram stain or standard-culture. Covered by macrolide, doxycycline, or fluoroquinolone.

**BOARD TRAP**

Like all atypicals it is beta-lactam resistant — the cell-wall-active drugs won't touch it.

## 8. Legionella spp.

**WHAT YOU SEE**

Severe-CAP/ICU exhibit, hotel/cruise icon

**WHAT IT ENCODES**

Severe CAP after hotel/cruise/water-aerosol exposure. Hyponatremia, GI symptoms/diarrhea, relative bradycardia, high LDH/ferritin. Diagnosed by urinary antigen (serogroup 1) and BCYE charcoal-yeast agar. Treat with fluoroquinolone or azithromycin.

**BOARD TRAP**

Hyponatremia + diarrhea + recent travel = Legionella, NOT Mycoplasma; urinary antigen is the rapid test (Mycoplasma has no urinary antigen).

### 9. Respiratory viruses

**WHAT YOU SEE**

Influenza/RSV/coronavirus tag, atypical panel

**WHAT IT ENCODES**

Influenza, RSV, and coronaviruses cause 20-30% of CAP and predispose to secondary bacterial (S. aureus, pneumococcus) superinfection. Bilateral interstitial infiltrates. Influenza treated early with oseltamivir.

**BOARD TRAP**

A biphasic illness (viral prodrome then sudden worsening) signals bacterial superinfection — don't attribute the second deterioration to the original virus alone.

### 10. HCAP abandoned

**WHAT YOU SEE**

Crossed-out HCAP exhibit, lower-left

**WHAT IT ENCODES**

The HCAP category was retired because it over-predicted resistant organisms and drove unnecessary broad-spectrum antibiotics. Modern guidance: assess individual MRSA/Pseudomonas risk factors rather than labeling HCAP.

**BOARD TRAP**

Do NOT automatically broaden to MRSA/Pseudomonas coverage just because a patient had healthcare contact — assess individual risk factors instead.

### 11. CA-MRSA exhibit

**WHAT YOU SEE**

Red glowing special exhibit, necrotizing/cavitary

**WHAT IT ENCODES**

Community-acquired MRSA carries the SCCmec type IV and PVL (Panton-Valentine leukocidin) genes, producing necrotizing pneumonia, cavitary lesions, and gross hemoptysis, often after influenza. Treat with vancomycin or linezolid.

**BOARD TRAP**

PVL toxin + post-flu necrotizing/cavitary pneumonia = CA-MRSA — beta-lactams (including ceftriaxone) fail; you must add vancomycin or linezolid.

### 12. PVL toxin

**WHAT YOU SEE**

Panton-Valentine leukocidin label, syringe

**WHAT IT ENCODES**

PVL is a pore-forming exotoxin that lyses leukocytes and drives the cavitary, necrotizing, hemorrhagic picture of CA-MRSA pneumonia. Associated with rapid clinical deterioration in young, previously healthy patients.

**BOARD TRAP**

Cavitation in a healthy young adult post-influenza is PVL-positive CA-MRSA, not typical hospital MRSA — the toxin, not just methicillin resistance, drives severity.

### 13. SCCmec IV / mecA

**WHAT YOU SEE**

Genetics placard, resistant-to-beta-lactams

**WHAT IT ENCODES**

The mecA gene encodes PBP2a, giving resistance to all beta-lactams. SCCmec type IV is the smaller, mobile cassette typical of community MRSA. Confirms why oxacillin/cephalosporins fail.

**BOARD TRAP**

mecA/PBP2a resistance means no beta-lactam works, including the 5th-gen anti-MRSA cephalosporin question's distractors — choose vancomycin/linezolid.

### 14. CA-MRSA susceptibilities

**WHAT YOU SEE**

TMP-SMX/clindamycin/doxy/vanc/linezolid list

**WHAT IT ENCODES**

CA-MRSA is often susceptible to TMP-SMX, clindamycin, doxycycline, vancomycin, and linezolid. For necrotizing pneumonia, linezolid or vancomycin is preferred (linezolid suppresses toxin production).

**BOARD TRAP**

Don't use clindamycin alone for severe necrotizing CA-MRSA pneumonia; reserve oral agents (TMP-SMX/doxy) for skin disease, not life-threatening lung infection.